

Academic Qualifications			
Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	7.91/10
2020	UP Board(XII)	S.K.S.V.M.I.C Madhav Nagar Shahjahanpur	78.6%
2018	UP Board(X)	S.K.S.V.M.I.C Madhav Nagar Shahjahanpur	89%

Scholastic Achievements	
- Secured 1st Rank in Formula Bharat Virtuals 2023 in Electric Vehicle category among 10 national teams - Secured All India Rank 8476 in Joint Entrance Exam Advanced 2021 among the 1.5 Lakh shortlisted candidates	

Work Experience	
SURGE'23, IIT Kanpur Prof. Animangsu Ghatak Chemical Engineering (May'23 - Jul'23)	
Agenda	● Identification and optimization of ink for writing on reusable paper for blind individuals
Approach	● Measured Surface energy of fabric glue and 3D outliner ink using water and Nitromethane liquid ● Acquired fundamental of surface roughness and measured roughness of reusable paper using Profilometer ● Conducted experiments to measure the change in the shape of ink drops and lines with drying time
Results	● Estimated Surface energy of the fabric glue is 41.69 mj/m² (Polar: 38.78 mj/m ² + dispersive:2.9 mj/m ²) ● The reusable paper is hierarchically rough, H_{rms} value is 1.9 μm on scale of 1600 μm×1600 μm

Key Projects	
Deep learning in Chemical engineering Simutech Project Chemineers Society (Feb'23 - May'23)	
Objective	● To analyze different CNN architecture to classify laminar and turbulent flow through simulation images
Approach	● Developed a comprehensive dataset of simulated images of flow in a 2D pipe using Comsol ● Auto augmented the data to ensure sufficient quantity for model training using horizontal flip and rotation ● Implemented and analyzed Resnet50, VGG16, AlexNet, LeNet5 and MobileNet on simulation dataset ● Finalized LeNet-5 for classification for the given problem statement based upon accuracy of the model
Results	● Developed model capable of classifying laminar and turbulent images with an approximate accuracy of 0.96

Algorithmic graph theory Association of Computing Activities (ACA) IIT Kanpur (May'23-Jul'23)	
Objective	● To explore basics of graph theory from an algorithmic perspective and application in real-world problems
Description	● Executed BFS and DFS for graph traversal, Prim's and Kruskal's algorithms for Minimum Spanning Tree ● Performed Cycle detection in directed and undirected graph and Kahn's algorithms for Topological sort ● Applied Dijkstra, A* search and Bellman-Ford algorithms for finding the shortest paths in a graph

Introduction to Machine Learning Chemineers Society (Dec'22)	
Approach and Results	● Cleaned and visualized the labeled data of Air quality index(AQI) of an Italian city using different techniques ● Implemented weight optimization techniques such as Gradient Descent, Normal equations from scratch ● Performed Artificial Neural Network using Keras on AQI dataset and obtained MSE loss of 0.0003536

Computational Methods in Engineering Matlab Course Project(ES0208) (3rd Semester)	
Description	● Implemented Bisection, False-Position, Fixed-Point, Netwon-raphson, Secant methods for finding roots of non-linear equations and Muller and Bairstow method for finding roots of arbitrary polynomial ● Applied Gauss elimination and LU Decomposition methods of solving simultaneous linear equations ● Implemented Romberg Integration and Gauss-Legendre Quadrature for numerical integration

Technical Skills	
Programming Languages & Libraries: C++, Python, Matlab, SQL, Numpy, Pandas, Matplotlib, Tensorflow, Sklearn Software & Tools: Git, Solidworks, Ansys, Optimum Lap, Craftware, Notion, L^AT_EX	

Positions of Responsibility	
Software Head, IITK Motorsports (May'23 - Present)	
Leadership	● Leading a 3-tier team of 10+ team members to work towards building an autonomous F1tenth scale car and testing different algorithms in order to participate in F1Tenth Championships in 2024
Management	● Made realistic and achievable Team Goals for the tenure and Timeline and Budget based on it ● Defining and assigning work , tracking progress and addressing issues that arises in the process ● Managed Snt Pavilion, recruitment , Sub-system division and training of junior team members
Initiatives	● Introduced idea of software subsystem and defined goals of the subsystem in the team IITK Motorsports ● Conducted regular Brainstorming sessions to review the approach and progress for the competition ● Instilled a documentation strategy that organizes team resources for long-term benefits of the team
Impact	● Coordination in intra and inter subsystem is established well and learning process picked up pace

Extracurricular Activities	
Mentored 5 freshmen acclimate to IIT Kanpur's environment and helped them with academic & emotional issues (2022) Collaborated with 3-tier team to execute the 10 days long Freshers' Orientation for 1200+ students (2022) - Coordinated with heads and colleagues for successful conduction of Blood Donation Camp during Techkrit'23 (2023)	